

Islam Asous

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EDUCATION

Hunter College, City University of New York (CUNY)

New York, NY

Bachelor of Arts in Computer Science, Minor in Mathematics

expected Dec 2026

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Computer Theory, Database Management, Object-Oriented Programming, Programming Methods, Foundations & Practice of Modern AI Systems, Automation Engineering, Practical Web Development, iOS Development, Linear Algebra, Applied Statistics

EXPERIENCE

NYC Department of Transportation

Aug 2026 – Present

Software Engineer Intern

Remote

- Scoping and delivering a minimum viable product against a formal project scope and change-order process, translating stakeholder requirements into defined deliverables and timelines.
- Collecting and analyzing data to inform application requirements and track project progress.
- Communicating progress through written status reports and oral presentations to leadership, practicing stakeholder management and technical communication.

Project Sisyphus, Clemson University

June 2026 – Present

Software Engineer REU

Remote

- Debugged and maintained an authentication-tracking browser extension (*WXT, Vue 3, TypeScript, Dexie.js*) supporting NSF-funded research (Award #2521037, \$650K) into usability gaps and authentication-fatigue-driven password reuse, capturing real-world authentication metrics across enterprise and AI-crowdsourcing platforms.
- Built a real-time broadcast notification system from scratch (*Vue 3, Pinia, Supabase Realtime*) enabling admins to push instant announcements to authenticated dashboard users, enforcing role-based access control via Postgres Row-Level Security.
- Embedded *Claude Code* into an agentic full-stack development workflow spanning implementation, test authoring, cross-repository debugging (Supabase permissions, OAuth redirects, CI issues), and PR review: iterating on prompts and AI-generated output to accelerate feature delivery while maintaining 90%+ test coverage.

CUNY Tech Prep

May 2026 – Present

Data Science Fellow

New York, NY

- Selected for a competitive fellowship with students from across the 11 CUNY senior colleges where Fellows create end-to-end data science and AI projects using tools such as Python, SQL, Pandas, scikit-learn, and Hugging Face.
- Participate in weekly courses and learn industry best practices for exploratory data analysis (EDA), feature engineering, statistical analysis, data visualization, LLMs, RAGs, vector databases, and generative AI applications.

Hunter College Computer Science Department

Aug 2025 – Present

Undergraduate Teaching Assistant

New York, NY

- Supported 600+ students across Computer Architecture I & II, Computer Theory, and OOP/Data Structures & Algorithms by leading weekly review sessions and office hours.
- Co-designed quizzes, midterms, and finals for courses of 150–300+ students; reduced grading turnaround by building automated evaluation tools with faculty.

PROJECTS

AI Agent From Scratch, Tool-Calling Research Agent | *Python, LangChain, Anthropic Claude API, Pydantic*

- Engineered a tool-calling research agent using LangChain and the Claude API that autonomously decides whether to invoke a live web-search tool or answer directly based on the query.
- Built a structured-output layer with Pydantic to parse raw model responses into a validated schema (topic, summary, sources, tools used) diagnosing a prompt-adherence failure and iterating on system instructions to enforce reliable, parser-compatible output across both tool-invoking and direct-answer queries.

RAG From Scratch, Local RAG Pipeline | *Python, Ollama, Llama 3.2, BGE Embeddings, Streamlit*

- Engineered a retrieval-augmented generation pipeline from first principles, implementing custom embedding storage, cosine similarity search, and prompt construction without relying on frameworks like LangChain.
- Built an in-memory vector database using a locally-hosted BGE embedding model and developed a Streamlit chat interface with multi-turn conversations, document uploads, and an inspectable retrieved-context view by iterating on prompt design to improve answer grounding.

For the Culture | *React, Vite, Express.js, Google Generative AI, Firebase*

- Expanded access to real-time food, health, and transit resources across all 5 boroughs in 8 languages by co-building and deploying a full-stack multilingual platform with a 4-person team, integrating a generative AI model for on-demand translation.
- Owned the React/Vite frontend architecture, including an interactive map consuming live NYC Open Data REST endpoints with real-time filtering by borough and resource type.

LEADERSHIP

Math Association of Hunter College

Aug 2024 – Aug 2026

President

New York, NY

TECHNICAL SKILLS

Programming: Python3, TypeScript, JavaScript, C++, Rust, Swift/SwiftUI, Dart, HTML/CSS, SQL (PostgreSQL)

Frameworks, Cloud & Tools: React, Node.js, Express.js, Flutter, Firebase, Supabase, Docker, REST APIs, Git/GitHub

Certifications: AWS Skill Builder - Developer, Cloud Operator (SysOps), DevOps (In Progress); CodePath AI201: Applications of AI Engineering (In Progress); Databricks Generative AI Fundamentals (In Progress)